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SMART INDIA HACKATHON 2026

Smriti Mitra

- **Problem Statement ID** – SIH26003
- **Problem Statement Title-** AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID-**
- **Team Name-** V-Max

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IDEA TITLE

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TECHNICAL APPROACH

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Tech Stack

- **Flutter** – Cross-platform Mobile App
- **Firebase** – Authentication & Firestore
- **TensorFlow Lite** – On-device AI Inference
- **Gemini AI** – Multilingual Conversational AI
- **Supabase** – Backend Services & Auth (Optional)
- **Node.js + Express** – REST APIs & Server Logic
- **SQLite** – Local Storage for Patient Data
- **Speech-to-Text** – Speech Recognition
- **Text-to-Speech** – Voice Guidance & Audio Feedback
- **Figma** – UI/UX Design & Prototyping

- **Flutter** – Cross-platform mobile application with an elderly-friendly interface.
- **Firebase** – Authentication, real-time database (Firestore) and cloud services.
- **TensorFlow Lite** – On-device AI inference for cognitive assessment.
- **Gemini AI** – Multilingual conversational assistant.
- **SQLite** – Local storage for patient data and reminders.
- **Speech Recognition + Text-to-Speech** – For voice-based interaction.
- **Secure REST APIs** – Encrypted data sync and secure communication.

Architecture Diagram

Patient Mobile App (Android / iOS)	Games, Assessment, Reminders, Voice/Chat Assistant, Emergency Support
↓ **Accessibility Layer**	Text size, contrast mode, voice guidance, multi-language support
↓ **AI Assessment Engine**	Cognitive, Speech & Behavioral Analysis
↓ **Cognitive Recommendation Engine**	Generates personalized games and activities
↓ **Brain Games Module**	Memory, Attention, Pattern Recognition, Daily Routine Recall
↓ **Progress Analytics**	Tracks performance, engagement and improvement
↓ **Caregiver Dashboard**	View reports, activity logs, reminders and alerts
↓ **Emergency & Reminder Service**	Medicine reminders, hydration alerts, Emergency SOS
↓ **Firebase Cloud Database**	Secure storage, real-time sync, encrypted data

AI Personalization Pipeline

Assessment Score → Cognitive Profile → Recommended Games → Adaptive Difficulty

Continuous Learning | Personalized Experience | Better Cognitive Outcomes

AI Modules

AI Dementia Assessment
Cognitive, Speech & Behavioral Analysis

- Evaluates memory, orientation, problem-solving
- AI-based speech analysis for fluency and response time
- Behavioral assessment for attention and daily functioning
- Generates cognitive profile and dementia severity level

Voice & Chat Assistant
Multilingual NLP and Voice Guidance

- Natural language interaction in regional languages (Assamese, Khasi, Manipuri, Mizo, Bodo, Garo, Nagamese, Hindi, English)
- Voice-guided navigation and instructions
- Answers queries about medicines, appointments and daily activities

Adaptive Gaming Engine
Personalized Difficulty using User Performance

- AI-driven game recommendations
- Dynamic difficulty adjustment based on accuracy, response time and engagement
- Culturally familiar games with North-East Indian content
- Improves memory, attention, reasoning and daily living skills

Caregiver Monitoring
Progress Dashboard, Reminders and Emergency Alerts

- Real-time progress tracking and activity reports
- Monitor assessment results and game performance
- Medicine and hydration reminders
- Emergency SOS alerts to family members or caregivers

Caring Minds • Stronger Communities • A Healthier North-East

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FEASIBILITY AND VIABILITY

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Technical Feasibility

Proven Tech Stack
Built using widely adopted technologies (Flutter, Firebase, TensorFlow Lite, Speech APIs, Gemini AI) ensuring smooth development and deployment.

AI Models Ready
Uses pre-trained and fine-tuned models for speech analysis, NLP, and game personalization.

Modular Architecture
Separate modules for assessment, games, reminders, and caregiver dashboard enable parallel development and easy updates.

Offline Capability
Core features like games, reminders and cached content work offline, ensuring usability in low-connectivity areas.

Secure & Compliant
Implements data encryption, secure authentication, and follows Indian data privacy guidelines (DPDP Act 2023).

System Overview

- Mobile App (Patient)
- Caregiver Portal (Web/Mobile)

Backend Services (Firebase)
- Authentication
- Database
- APIs
- Cloud Functions

- AI & ML Models (Assessment, Personalization, NLP, Speech)
- Secure Storage (Encrypted Data)

Feasibility Flow

Step 1
Data Collection
(Game responses, speech input, user activity, caregiver data)

Step 2
AI Processing
(Cognitive, Speech & Behavioral Analysis)

Step 3
Personalization
(Recommend games, set difficulty, generate reminders)

Step 4
Delivery on Mobile App
(Offline/Online)

Step 5
Continuous Learning
(Performance tracking & model improvement)

Improves Accuracy | Adapts to User Needs | Scales Across Regions

Operational Viability

Elder-Friendly
Simple navigation, large buttons, voice guidance, and minimal cognitive load.

Remote-Ready
Works in low-connectivity areas with offline access to key features.

Caregiver Integration
Enables family members to monitor progress, reminders, and emergency alerts.

Scalable
Can be deployed across North East India with support for multiple regional languages.

Cost-Effective
Uses open-source tools and cloud services, keeping implementation and maintenance costs low.

Sustainable
Modular design allows easy updates, new games, and integration with future healthcare services.

Real-World Usage Scenario

Patient uses SmritiMitra at home → Caregiver monitors progress remotely → Doctor provides guidance when needed

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IMPACT AND BENEFITS

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SIH

Impact

Bridging the Care Gap
Helps address the limited access to cognitive therapy and neurologists in remote and rural areas of North-East India.

Better Cognitive Health
Enables early assessment and continuous cognitive training, helping slow down cognitive decline.

Empowered Caregivers
Provides tools and insights for families to monitor, support, and care for elderly patients even from a distance.

Culturally Inclusive Care
Uses familiar regional content, languages, and local contexts to make dementia care more relatable and effective.

Scalable Public Health Solution
Can be deployed widely across North-East India to support a large elderly population with minimal infrastructure.

Improved Quality of Life
Promotes mental engagement, independence, and emotional well-being for elderly users and their families.

Future Scope

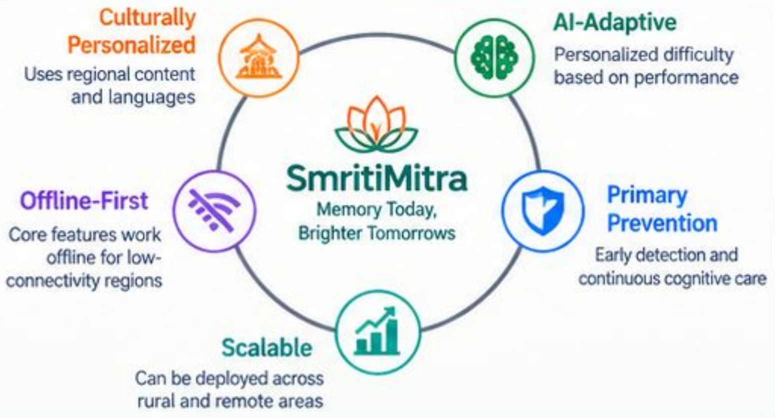
Wearable Integration
Sync with smartwatches/health bands to track heart rate, sleep patterns and activity for deeper cognitive insights.

Smart Home Integration
Voice-enabled smart speakers for reminders, safety alerts, and hands-free interaction.

Expanded Regional Coverage
Add more North Eastern languages and culturally relevant content.

Clinical & Institutional Adoption
Collaboration with hospitals, ASHA workers and eldercare institutions for large-scale deployment and real-world validation.

Key Advantages



Benefits

Business Model & Sustainability

Government Partnerships (B2G)
Collaboration with state health departments for deployment across North Eastern states.

Institutional Licensing (B2B)
Partnerships with hospitals, eldercare centers, and NGOs for group access and monitoring.

Freemium Model (B2C)
Core features free for patients; premium features like advanced analytics, extended game library, and cloud backup for families and caregivers.

Sustainable Model
Low-cost, scalable, and designed for long-term impact through government support, institutional adoption, and community outreach.

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RESEARCH AND REFERENCES

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Key Research and References

1. **AI4Bharat. IndicTrans2: A Multilingual Neural Machine Translation Model for Indian Languages, 2023.**
https://ai4bharat.iitm.ac.in
Open-source multilingual translation research supporting regional language integration for North-East Indian languages.
Supports multilingual voice and text interaction in SmritiMitra.

2. **World Health Organization (WHO). Global Action Plan on the Public Health Response to Dementia 2017–2025.**
https://www.who.int
Guides global strategies for dementia care, early detection, and community-based support.
Provides global best practices for dementia care and policy guidance.

3. **GBD 2019 Dementia Forecasting Collaborators. The Lancet Public Health, 2022.**
https://www.thelancet.com
Provides epidemiological baseline and future projections of dementia burden, including data relevant to India.
Highlights the growing need for early intervention and scalable solutions.

4. **MoHFW, Govt. of India. National Programme for Health Care of the Elderly (NPHCE) Guidelines.**
https://dohfw.gov.in
Policy framework for elderly healthcare, including roles of caregivers, ASHA workers, and community health support.
Aligns our solution with national elderly healthcare initiatives.

5. **MeitY, Govt. of India. Digital Personal Data Protection Act, 2023.**
https://www.meity.gov.in
Provides the legal framework for data privacy, user consent, and secure handling of personal data.
Guides our privacy and data security architecture (AES-256).

6. **Regional Studies and Cultural Resources (NER)**
Various government publications, cultural archives, and local knowledge sources.
Used to include culturally familiar content, symbols, and languages from North Eastern India to improve engagement and relevance.
Helps design context-aware games and content for NER users

Prototype Links

Project Repository (GitHub)
https://github.com/sparshagarwal0411/dimentia-games
Source code, documentation and setup guide.

Live Demo / Website
https://dimentia-games.vercel.app/
Try the working prototype.

Key Organizations Referenced

World Health Organization (WHO)
Global dementia care guidelines

THE LANCET
The Lancet Public Health
Epidemiological research and projections

Ministry of Health and Family Welfare (MoHFW), Govt. of India
NPHCE guidelines for elderly healthcare

MeitY
Ministry of Electronics and IT (MeitY)
Digital Personal Data Protection Act, 2023

AI4Bharat
AI4Bharat
Open-source multilingual AI models

> *"Our solution is built on credible research, government guidelines, and real-world needs — adapted for the unique context of North East India."*
> — SmritiMitra Team